

**ECSE-4040 DIGITAL ELECTRONICS
MID-TERM EXAM, SPRING 2026
RENSSELAER POLYTECHNIC INSTITUTE**

ASSIGNED: W, FEBRUARY 11; DUE (AT BEGINNING OF STUDIO): W, MARCH 11

STATEMENT OF UNDERSTANDING

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"I understand that this Exam is officially administered under an Academic Honor Code: I hereby state that I have not used any other source material in completing this Exam other than my personal notes, the course notes, the course text, and/or SPICE software and documentation. In addition, I have not consulted or interacted, in any way or form, with anyone else, or any generative AI tool (such as ChatGPT), except, possibly, the Course Instructor, in completing this Exam."

Name: _____ (printed) Signature: _____

Student Number: _____ Date: _____

(Submit this signed page along with your work on LMS)

Note, for all SPICE verifications, give at least (1) Schematic, and (2) Plot or, if appropriate, Operating Point Data.

There are three problems on this exam. Use the given 130nm process parameters below for hand calculation for all problems unless the problem explicitly states these are not to be used:

Below is a table of process parameters for a generic 0.13 μm technology.

0.13 μm Process technology parameters

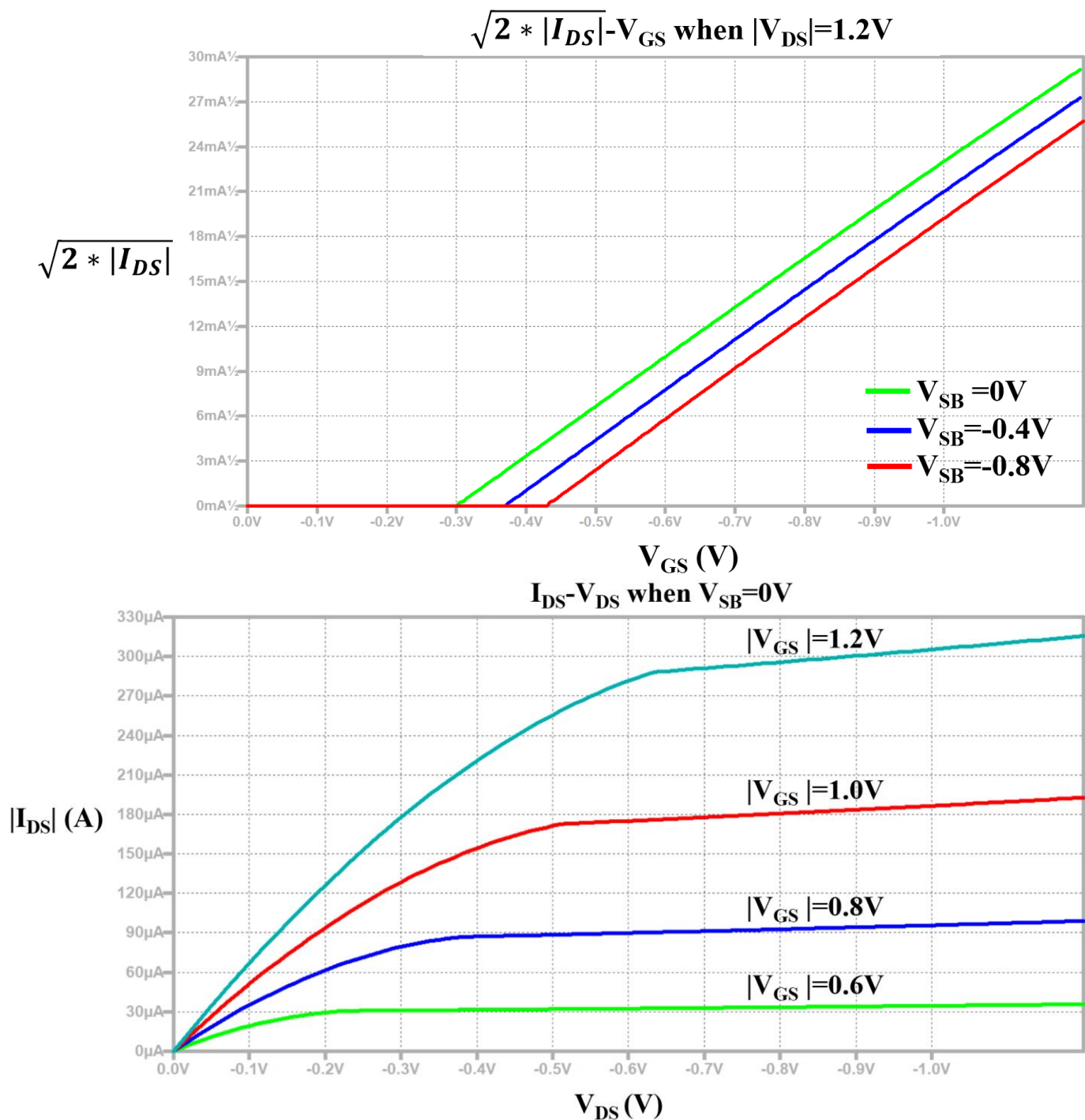
Typical Technology Parameters	Symbol	NMOS	PMOS
Gate thickness	t_{ox} [Å]	22	22
Poly doping level	N_D [cm^{-3}]	3×10^{20}	—
Poly doping level	N_A [cm^{-3}]	—	3×10^{20}
Substrate doping level	N_A [cm^{-3}]	3×10^{17}	—
N-well doping level	N_D [cm^{-3}]	—	3×10^{17}
Number of surface state charges	N_{ss} [cm^{-2}]	6×10^{11}	6×10^{11}
Electron Mobility	μ_n [cm^2/Vs]	270	—
Hole Mobility	μ_p [cm^2/Vs]	—	70
Critical Field x Length	$E_C L$ [V]	0.6	2.4
Saturation velocity	v_{sat} [cm/s]	8×10^6	8×10^6
Gamma	γ [$\text{V}^{1/2}$]	0.2	0.2
Oxide capacitance	C_{ox} [F/cm^2]	1.6×10^{-6}	1.6×10^{-6}
Threshold voltage	V_T [V]	0.4	− 0.4
2 x Fermi voltage	$2 \phi_F $ [V]	0.88	0.88
Channel length modulation	λ [V^{-1}]	0.2	0.2

Problem 1 (20 pts. HAND, 10 pts. SPICE):

For this question, the following terms differ from the given 130nm process parameters. All other terms are assumed to be the values as listed in the given table above.

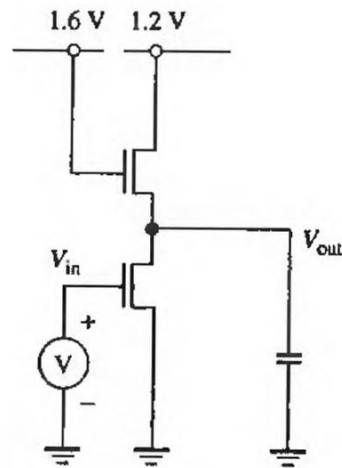
- L (here given as 130nm)
- μ_p
- V_{T0}
- γ
- λ
- E_c, E_{cL}

For this modified process, the following two plots are given for a $W/L = 10$ PMOS transistor:



- a) By HAND analysis using the plots above, find V_{T0} , E_{CL} , γ , and λ assuming they are **NOT** the same as the HJS textbook 130nm process values. Use **short channel MOSFET equations**.
 - i. State and verify each transistor-state assumption in hand calculations.
 - ii. Find μ_P for extra credit (Hint: where is k' ?)!
- b) Verify results for Part (a) using SPICE.
 - i. Use the provided **MOS2.txt** model file on LMS.
 - ii. This file has the default HJS 130nm process values. Enter your calculated values and perform DC-sweep simulations to verify that your transistor matches the given transistor.
 - i. Enter W and L as normal in the NMOS symbol in the schematic.
 - ii. Modify the MOS2.txt file with your modified process values where:
 1. $vt0 = V_{T0}$
 2. $\gamma = \gamma$
 3. $\lambda = \lambda$
 4. $U_0 = \mu_P$
 5. E_{CL} is calculated from V_{max} (v_{sat}) and L, and need not be entered.
 - iii. Show both $(2*|I_{DS}|)^{1/2} - V_{GS}$ and $|I_{DS}| - V_{DS}$ plots like above to verify your parameters are correct.
 - i. **Don't use the same color scheme as the plots above so that we know you ran the simulation yourself.**
 - ii. Modify parameters as needed to best match the provided plots.
 - iv. To avoid a bug in LTspice, set your V_{T0} to not be a value that is divisible by 0.01 (Ex: use 0.401V instead of 0.4V).

Problem 2 (20 pts. HAND, 10 pts. SPICE):



Given:

$L_N = L_P = 100\text{nm}$, $V_{DD} = 1.2\text{V}$.

Perform the following tasks:

(a) Determine the ratio of W_D/W_L for the following cases:

(i) $V_{out} = V_{OL} = 0.1\text{V}$,

(ii) $V_{out} = V_{TN} = 0.4\text{V}$,

(iii) $V_{out} = V_S = 0.6\text{V}$.

(b) Verify results for Part (a) using LTSPICE. Explain any discrepancies with the HAND analysis.

For the enhancement NMOS/PMOS, use the NMOS/PMOS model in **MOS2.txt**.

Problem 3 (25 pts. HAND, 15 pts. SPICE):

Given parameters:

$L = 100\text{nm}$, $C_L = 75\text{fF}$, $V_{DD} = 1.2\text{V}$.

- (a) Design a CMOS gate whose output is $\text{Out} = AB + \bar{B}C$. Calculate the width of the transistors so that $t_{PHL} = t_{PLH} < 50\text{ ps}$.
- (b) Verify results for Part (a) using LTSPICE. Explain any discrepancies with the HAND analysis.

Submit your work on LMS under **Midterm**

- Show your work.
 - Include images of your hand calculations with work and final results.
 - Include screenshots of LTspice with all circuit schematics with all output plots, clearly showing all V_{OH} and V_{OL} values.
- Due date: Wednesday, March 11 at 9 am.